

PT100 four-wire RTD readout

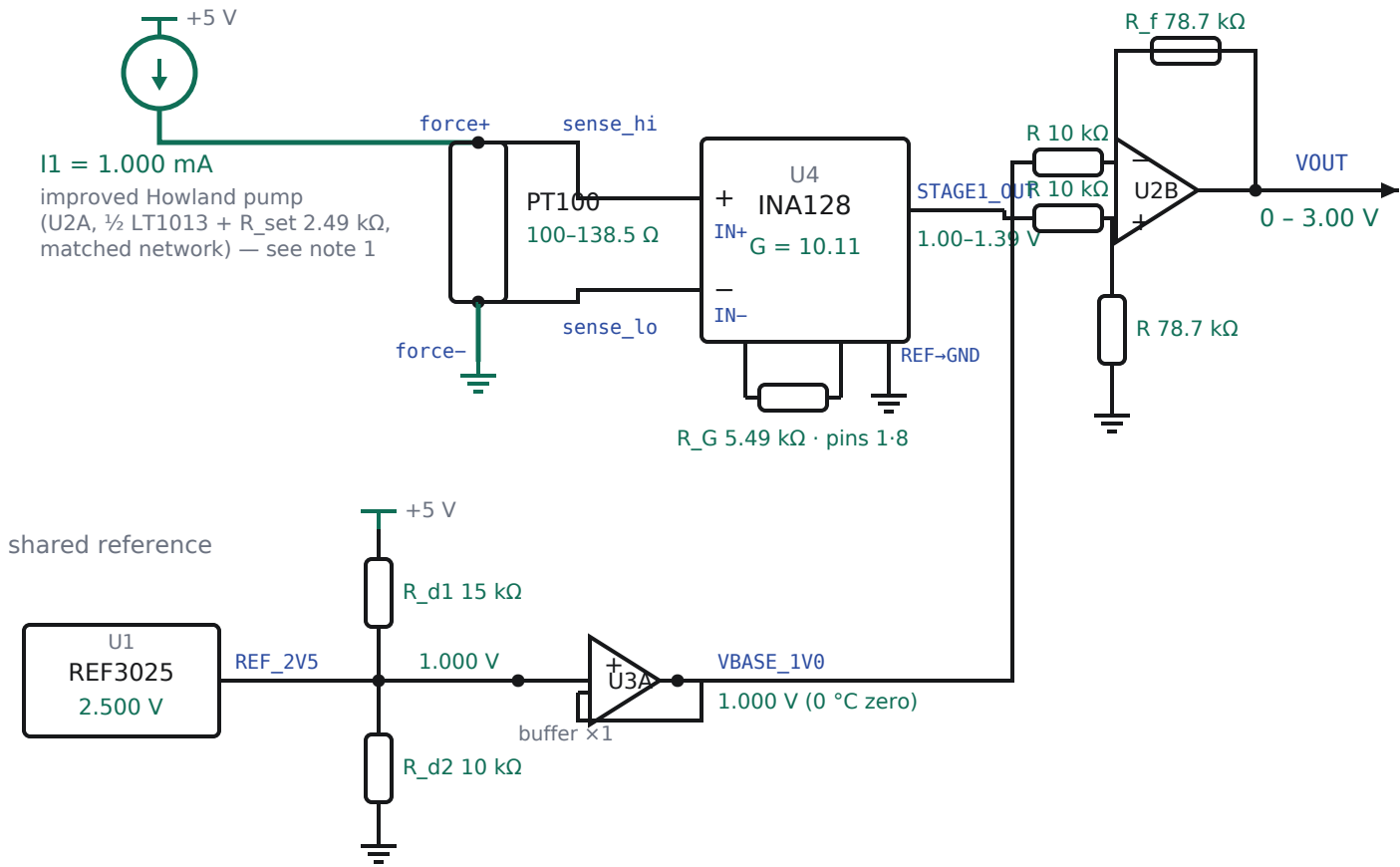
constant-current excitation · Kelvin (4-wire) sensing · instrumentation + difference amplifier

excitation

4-wire PT100

stage 1 · INA128 ×10

stage 2 · difference amp ×7.87



bill of materials

Ref	Part	Value / role
U1	REF3025	2.500 V, 25 ppm/°C
U2	LT1013	dual: Howland + diff amp
U3	LT1013	dual: baseline buffer +spare
U4	INA128	instrumentation amp ×10
R_set	2.49 kΩ	0.1% — sets I=Vref/R_set
R_H1-4	10 kΩ	0.1% matched (Howland)
R_G	5.49 kΩ	0.1% — INA gain (pins 1-8)
R_in ×2	10.0 kΩ	0.1% — diff-amp inputs
R_f ×2	78.7 kΩ	0.1% — diff-amp feedback
R_d1	15 kΩ	0.1% — baseline divider
R_d2	10 kΩ	0.1% — baseline divider
RTD	PT100	resistor model in SPICE
C ×6	100 nF	decoupling / supply pin

- excitation current (force leads)
- signal / sense (no current → Kelvin)

note 1: current source realized as improved Howland pump (TI SBOA437); absolute current is non-critical — the baseline shares REF3025, so first-order reference drift cancels.

note 2: R_d1/R_d2 set the 0.400 ratio → 1.000 V; the buffered baseline tracks the reference.

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Portfolio Project 5	Sheet 1 / 1
Rev A · 2026-07-06	Supply: ±5 V